

Assignment 2: Preliminary Results

8/10/2025

Attempt 1



In Progress

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3 Attempts Allowed

Details

Objective

Build and evaluate **preliminary machine learning models** based on the data processing steps and feature engineering outlined in your Analytic Plan (Milestone 1). Identify which models perform well, diagnose areas for improvement, and plan next steps for the Final Report.

Instructions

1. Data Preparation & Feature Implementation

- Carry out the data processing procedures proposed in Milestone 1 (cleaning, encoding, transformation, etc.).
- Create and integrate the new features outlined in your Analytic Plan (e.g., aggregated or engineered variables).

2. Model Building

- Implement multiple supervised learning models (e.g., Logistic Regression, kNN, Naive Bayes, LDA).
- Compare each model's predictive performance on a held-out test set or through cross-validation.
- Include a comparison against the null model (baseline) to gauge relative improvements.**

3. Performance Evaluation

- Evaluate each model using key metrics: **accuracy, AUC, precision, recall**.
- Generate and discuss visualizations (e.g., **ROC curves, confusion matrices**) to clarify performance.
- Explain **why** different models might yield different results (e.g., algorithmic differences, data assumptions).

4. Variable Selection & Insights

- Refine your initial variable selection using **robust methods** (correlation analysis, forward/backward, stepwise).
- Identify which variables are consistently important, and note any that are irrelevant or redundant.
- Discuss **how variable selection** affected model performance and what that implies for business insights.

5. Recommendations for Final Report

- Summarize **further steps** needed for Phase 3 (e.g., advanced feature engineering, hyperparameter tuning, new algorithms like Random Forest).
- Suggest additional EDA angles or data sources that might enhance predictive performance.

Deliverable

1. Model Implementation & Results

- Train and evaluate at least two or three models from your initial shortlist.
- Provide key metrics (accuracy, AUC, precision, recall) and relevant visualizations (ROC curves, confusion matrices).

2. Robust Variable Selection Process

- Document the **methods** (e.g., stepwise selection) used to refine your predictors.
- Indicate which variables remain critical vs. which were dropped.
- Include **discussion** of any notable changes in performance after refining variables.

3. Model Comparison

- Present a **summary table** of performance metrics for each model.
- Offer a **qualitative comparison** (e.g., interpretability vs. accuracy trade-offs).
- Explain **why** certain models may perform better or worse on your dataset.

4. Recommendations for Phase 3

- Outline next steps and **improvements** (e.g., feature engineering ideas, hyperparameter tuning, advanced ensemble methods).
- Identify potential **new data** elements or EDA directions to explore further.

Formatting and Submission

- Length: Up to 10 single-spaced pages (excluding appendices).
- Format: MS Word or PDF (e.g., from RMarkdown), 12-point Times New Roman font.

• Submission:

1. Upload your file to Canvas (either by selecting Upload or dragging and dropping the file).
2. Click Submit Assignment to finalize.

Important:

- Include **only** the charts/graphs and analytic outputs that support your findings and discussion.
- Excluding necessary evidence or including irrelevant output will adversely affect your grade.
- Continue to be **thoughtful and purposeful** in your analysis and presentation.

View Rubric

Class Project Milestone 2 Rubric

Criteria	Ratings		Points
Model Implementation & Results view longer description	Full Marks 35 pts	No Marks 0 pts	/35 pts
Refined Data Preparation & Feature Engineering view longer description	Full Marks 20 pts	No Marks 0 pts	/20 pts
Variable Selection view longer description	Full Marks 20 pts	No Marks 0 pts	/20 pts
Model Comparison view longer description	Full Marks 15 pts	No Marks 0 pts	/15 pts
Recommendations & Organization view longer description	Full Marks 10 pts	No Marks 0 pts	/10 pts

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